Jacob Horne

(669) 261-1669 | jacobhorne.jth@gmail.com | linkedin.com/in/jacobhornejth | github.com/jacobhorne-jth | Portfolio | U.S. Citizen

Education

University of California, Irvine June 2028
Bachelor of Science in Computer Science; Campuswide Honors Collegium *GPA: 3.92*
Coursework: Data Structures & Algorithms, Operating Systems, Distributed Systems, System Design, ML & Data Mining Artificial Intelligence, Databases, Computer Architecture, Software Engineering, Computer Vision, Probability & Statistics

Experience

Sandia National Laboratories Aug 2026 – Present
Software Engineer Intern *Livermore, CA*
- Building full-stack applications and backend services that make Sandia AI research accessible through user-facing tools

Capital One Jun 2026 – Aug 2026
Software Engineer Intern *New York, NY*
- Built Databricks sampler over ∼**800M daily customer-response outcomes**, enabling **28× faster** tests across longer history
- Automated analyst tool for testing new campaigns and ranking models with Airflow: joins, sampling, **18 PySpark checks**
- Replaced production ML-vs-random routing API with SDK; prevented timeouts, cut latency **25% for 60M+ users/month**
- Added unit and end-to-end SDK tests, validating stable assignment and downstream message/deal rankings

UCI Digital Learning Lab Dec 2025 – Present
Research Assistant *Irvine, CA*
- Evaluated Advocate/Skeptic reasoning on ASAP dataset essays using LLM-as-judge rubrics across four writing traits
- Analyzed token log-probs with NumPy/Pandas/Matplotlib to relate confidence features to rubric-based reasoning quality
- Co-authored **3 publications** on LLM confidence and multi-agent reasoning at **ICLR Workshop**, **ACL SRW**, **GEM** (2026)
- Investigating token-level confidence across multi-turn text-to-SQL query generation, execution, and iterative repair cycles

California Institute for Telecommunications and Information Technology Apr 2026 – Present
AI/ML Undergraduate Researcher *Irvine, CA*
- Developed PyTorch/GRU models replacing **hours-long** heat-transfer FEA simulations for wall-temperature prediction
- Extended a GRU into a physics-informed neural network constrained by heat-transfer equations and initial conditions
- Achieved **12,000× faster inference than heat-transfer simulations** at scale with $\mathbf{R^2 = 0.99}$ **and MAE = 1.54°C**

Commit the Change at UCI Oct 2025 – Jun 2026
Full-Stack Developer *Irvine, CA*
- Built React/TypeScript scheduling/quota platform with quota creation and staff access, supporting **8,000+ patients/year**
- Developed Node.js/Express.js/PostgreSQL provider directory for **24+ providers**, saving **12+ staff hrs/week**

SENS Psychology Sep 2025 – Jan 2026
Software Development Intern *Arlington, VA*
- Built Next.js/TypeScript intake and ticketing platform used by **30+ staff across 1,200+ monthly patient interactions**
- Cut **patient intake time 50% and manual emails 75%** via Node.js/PostgreSQL automation and Azure notifications

Projects

Agonus - AI Agent Trading Platform | *Blockchain at UCI* Oct 2025 – Apr 2026
- Architected LangChain ReAct agents (GPT-4o-mini) with tools for market data, trade execution, planning, and memory
- Designed Celery/Redis scheduling, locking, and crash recovery to reliably orchestrate **10+ concurrent autonomous agents**
- Contributed to decentralized AI trading platform with on-chain tournaments, live leaderboards, and token rewards

Prop.Intel - AI Property Risk Platform | *IrvineHacks 2026* Feb 2026 – Mar 2026
- Built AI property-risk platform using ATTOM/FEMA data to generate real-time disaster-risk reports for U.S. properties
- Developed ridge regression model across **3,100+ counties and 8 hazard types** to quantify nationwide disaster exposure
- Built interactive property-risk dashboards for ownership timelines, hazard breakdowns, and AI-generated risk summaries

Technical Skills

Languages: Python, C++, Java, TypeScript, JavaScript, SQL, C, R, MIPS Assembly
Backend/Web: React, Next.js, Node.js, Express.js, FastAPI, REST APIs, PostgreSQL, Redis, Celery
Data/ML: PyTorch, scikit-learn, PySpark, Databricks, Airflow, TensorFlow, NumPy, Pandas, Matplotlib, LangChain, LangGraph
Systems/Cloud: Linux, Bash, Docker, Git, CI/CD, CUDA, AWS, ECS, Fargate, S3, CloudWatch, Azure